Companion Note to "The Unique Games Theorem: A Short Self-Contained Proof": Attribution, Provenance and Checks

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This note accompanies the paper *The Unique Games Theorem: A Short Self-Contained Proof* [1] and is deposited with it at doi:10.5281/zenodo.23265572. The paper gives a short proof of the theorem of OpenAI's preprint *The Unique Games Theorem* [2]; a longer presentation that follows the source closely is [3]. Below we list the paper's departures from the source, credit the contributors to each simplification, describe how the paper was reviewed, and report a run of the Comparator check on OpenAI's Lean formalization. Section, theorem, proposition and lemma numbers refer to the paper.

Attribution and changes of presentation

The construction and main proof are from [2]. The changes below simplify the presentation and some proof steps; none is a claim of a new theorem or an independent verification of the preprint.

Parity equations, tables and weights (Sections 2 and 5). The source makes the variables of each equation distinct by replacing every variable with 100 clones and listing all distinct-clone choices with conditional weights; its bound uses sign rounding of clone biases and conditioning. We instead cancel repeated variables and replace each unary equation by three equations on a triangle of fresh variables; the bound $\frac{2}{3} + \frac{1}{3}\,\mathrm{OPT}$ suffices, since only a constant gap is needed, and turns the source's constants 4/5, 14/15 and 3600 into 6/7, 20/21 and 7056. The source rounds occurrence weights before subdividing; we keep exact multiplicities, using the equally likely random strings of Håstad's verifier. We define a key as the coarsest factorization of a table, so pullback compatibility follows from uniqueness; the source checks that explicit third-bit intercept corrections cancel.

The noise gadget (Section 3). The source builds the recursion on translation spaces, starts the gadget's rank tracking from a generic space, and realizes its output by a quotient, then copies indexed by injections and a second quotient. In the paper the recursion is written in direct coordinates $V_t = W_t \oplus B_{\mathrm{blk}}$, so the output is already a subspace and no quotient is needed; genericity needs only the alignment bound, without excluding proportional pairs; instead of following one lift (in the source's sense) through chosen sections, we pull back the whole observer family; and instead of following the random leaf path with a filtration and telescoping the expected drops, we prove a detection bound for every family by induction on the height, whose potential $H_r - \theta N_S/r$ also admits an arbitrary starting space at a one-time cost below $2^{r_0}\theta$. The threshold becomes r_0, and only output dimensions divisible by 3 are realized,

which suffices for every later use. In the kernel identity and the genericity argument, squaring the alignment equation and formal differentiation over $\mathbb{F}_2(y_{ij})$ replace the source's square-root monomial-independence and double-annihilator step, and one annihilating row vector replaces its nonzero augmented minor when passing to F; the closing point-indicator count follows the source. The source's update space $\{(a, Q(a)+w) : a \in A,\ v \cdot w = 0\}$ is written in the paper, via $w \mapsto w + t^2 Q(v)$ at $a = tv$, as the kernel of one F-valued equation, which yields its dimension, the update count and the kernel identity.

Soundness (Section 4). The source's matrix-chart argument credits Barak, Kothari and Steurer [4]; the needed calculation is proved in the paper, in the proof of Lemma 3. The source shows that good row advice is frequent by recoloring randomly inside K and applying a concentration and counting bound; we instead bound acceptance directly by η plus twice the measure of the union of good row fibers, which needs neither. The source selects useful projected data and decodes affine targets $Mz+u$; we keep constant-output certificates, transfer a score odd along a random nonzero $h \in \ker A$, and let the second player guess the output and Fourier-sample one row along h, which yields a legal answer at every sampled frequency; Cauchy–Schwarz then gives unconditional agreement without revealing the certificate. The noise comparison likewise expands $\mathbf{1}_{F_U=u}$ rather than averaging over output characters. The source takes k along cubes so that $\beta = k^{-2/3}$ is rational; β enters only the soundness experiment, never the output game, so every large k is allowed in the paper. In that experiment, singleton columns of M_O are drawn from $\ker A$ and those of T_O are zero, so every singleton is clean; the source uses only singletons whose advice vanishes.

Provenance and review

Several drafts of the paper received source-based reviews by separate Claude instances and by Codex; Codex had also helped write an earlier draft. After reviewing the draft that became the longer presentation [3], Codex proposed, at the author's request and in eight rounds, most of the departures listed under Attribution. Claude proposed the triangle, which replaced Codex's seven-variable gadget, the keys as coarsest factorizations, the induction on the height, the update count, the single annihilator and the update space as one kernel in Section 3, the agreement bound through one conditional expectation in Section 4, the two-stage sampling in Section 4.3, and three restructurings: one soundness proposition (Proposition 2), one lemma from test acceptance to good row fibers (Lemma 3), and one choice of the gadget parameters in Section 3. The model gpt-6-astra (OpenAI) proposed the transferred correlation and a one-row decoder with a random character, and a separate Codex run through the API (gpt-6.1-sol, OpenAI) proposed a one-row decoder that guesses the output value instead. Muse Spark 1.3 (Meta) proposed compressing parts of Sections 2, 3, 4.3 and 5 and reading the update count through the dot product of the output change with v. Gemini 3.8 Flash (Google) proposed treating the score transfer and the decoder as one subsection. A separate Claude reviewer checked each round against the preceding draft, and its fixes are applied. These reviews do not certify the preprint or constitute formal verification of the paper, and no human expert has reviewed it.

Lean check

OpenAI's repository [2] contains a Lean 4 formalization of the source theorem, with a Comparator challenge that permits only the standard axioms. We report one run of this check, made on 9 October 2026 at the revision pinned in [2], as an unprivileged user on a four-core x86-64 cloud machine. It is the run reported in Appendix A of [3]. The check script

(`ugc_lean_check.py`) and its complete output logs (`ugc-lean-ugt-check.log` and the output directory, as `ugc_out.zip`) are deposited with that paper at doi:10.5281/zenodo.23264902.

Repository	`openai/math`, revision `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`
Tools	Lean 4.34.1 (pinned toolchain); Comparator commit `d03acab` (tag `v4.34.0`)
Command	`lake env comparator ComparatorChallenges/UniqueGamesTheorem.json`
Declaration	`OAI.UniqueGamesTheorem.theorem11`
Axioms	`propext`, `Classical.choice`, `Quot.sound`
Verdict	"Lean default kernel accepts the solution", "Your solution is okay!"

Procedure and result. Following the repository's instructions, our script ran `lake update` (manifest and toolchain unchanged), `lake exe cache get` and the command above in the directory `lean`, then a separate `lake build` of the solution and challenge modules and `#print axioms`. The challenge module states the declaration's type: `BinaryGapReduction` ε δ is nonempty for all real $\varepsilon, \delta \in (0, 1/2)$. Comparator exited with code 0, the build succeeded, and the solution's axioms include no `sorryAx`; only the challenge module uses `sorry`, by design, to state the type. The run took 66 minutes.

Caveats. The sandbox was not enforced: `landrun` started, but in our test a write outside its allowed directory succeeded, and the `systemd-run` guard recommended in Comparator's README was unavailable. The check therefore lacked the recommended isolation. We did not audit whether the formal statement matches Theorem 1 of the paper. Above all, the run concerns OpenAI's formal development of the source theorem. It does not check the paper's simplifications listed under Attribution, or anything else written in the paper, and the paper is not derived from the formalization.

References

[1] Ariel Elboim. *The Unique Games Theorem: A Short Self-Contained Proof*. 2026. Deposited with this note at doi:10.5281/zenodo.23265572.

[2] OpenAI. *The Unique Games Theorem*. Preprint, 23 September 2026, 58 pages. Pinned source revision: GitHub source and paper.

[3] Ariel Elboim. *A Compressed Presentation of OpenAI's Proof of the Unique Games Theorem*. Preprint, 2026, doi:10.5281/zenodo.23264902.

[4] Boaz Barak, Pravesh K. Kothari, and David Steurer. *Small-Set Expansion in Shortcode Graph and the 2-to-2 Conjecture*. Proceedings of ITCS, LIPIcs 124, 9:1–9:12, 2019.